

Brutal Series - Paper B55 (Soil | Exponents & Powers | Heat | Lines & Angles)

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A farmer pours 240 mL of water into a pipe on sandy soil and it fully soaks in 16 minutes; the same 240 mL on clayey soil takes 48 minutes. How many times faster is the sandy soil's percolation rate? [\[Soil\]](#)
 - (A) 2 times faster
 - (B) 3 times faster
 - (C) 1.5 times faster
 - (D) 16 times faster
 - (E) 4 times faster
 - (F) 6 times faster
 - (G) 32 times faster
 - (H) 48 times faster

2. Simplify the expression $(2^3 \times 2^4) / 2^5$ and then express the result as a power of 2. [\[Exponents & Powers\]](#)
 - (A) 2^{60}
 - (B) 2^1
 - (C) 2^7
 - (D) 2^{12}
 - (E) 2^3
 - (F) 2^2

3. A thermometer reads 35 degC. Using the conversion $F = (9/5)C + 32$, what is this temperature in degrees Fahrenheit? [\[Heat\]](#)
 - (A) 99 degF
 - (B) 105 degF
 - (C) 63 degF
 - (D) 95 degF
 - (E) 61 degF
 - (F) 67 degF
 - (G) 127 degF

4. Two angles are complementary. One angle is 18 degrees more than the other. Find the larger angle. [\[Lines & Angles\]](#)
 - (A) 48 degrees
 - (B) 60 degrees
 - (C) 45 degrees
 - (D) 72 degrees
 - (E) 81 degrees
 - (F) 54 degrees
 - (G) 36 degrees

5. In a soil profile the dark topmost horizon is 12 cm thick, the next horizon is 3 times as thick as the top one, and the third horizon is 18 cm more than the second. What is the total depth down to the bottom of the third horizon? [\[Soil\]](#)

- (A) 90 cm
- (B) 102 cm
- (C) 84 cm
- (D) 66 cm
- (E) 96 cm
- (F) 120 cm

6. Find the value of $(3^2 \times 3^3)$ and then subtract 5^2 from it. [\[Exponents & Powers\]](#)

- (A) 243
- (B) 200
- (C) 118
- (D) 218
- (E) 250
- (F) 18
- (G) 268

7. Water is heated from 18 degC to 78 degC. Each degree of rise needs 200 joules for this sample. How many KILOJOULES of heat are supplied? (1 kJ = 1000 J) [\[Heat\]](#)

- (A) 120 kilojoules
- (B) 1.2 kilojoules
- (C) 12000 kilojoules
- (D) 60 kilojoules
- (E) 15.6 kilojoules
- (F) 12 kilojoules

8. Two angles are supplementary and are in the ratio 4 : 5. Find the SMALLER angle. [\[Lines & Angles\]](#)

- (A) 45 degrees
- (B) 40 degrees
- (C) 80 degrees
- (D) 100 degrees
- (E) 90 degrees
- (F) 72 degrees
- (G) 20 degrees

9. Tests show 100 g of dry clayey soil can hold 45 g of water, while 100 g of dry sandy soil holds only 18 g. For 250 g of each dry soil, how much MORE water does the clayey soil hold than the sandy soil? [\[Soil\]](#)

- (A) 45 g
- (B) 112.5 g
- (C) 67.5
- (D) 90 g
- (E) 67.5 g
- (F) 157.5 g

10. A number is written in standard (scientific) form as 4.6×10^5 . What is this number written in full (ordinary form)? [\[Exponents & Powers\]](#)

- (A) 46000
- (B) 4600
- (C) 4.600000
- (D) 460000
- (E) 4600000
- (F) 406000
- (G) 45000

11. Three metal rods - copper, steel and glass - are heated equally at one end. Wax stuck on the far ends melts in this order: copper first, steel next, glass last. What does this reveal, and which is the best conductor? [\[Heat\]](#)

- (A) Glass is the best conductor because it melted last
- (B) The wax melts by chance, telling us nothing
- (C) Heat conduction speed differs; copper is the best conductor
- (D) Steel is best because it is in the middle
- (E) All conduct equally; glass is best
- (F) Copper is the worst conductor

12. Two lines cross at a point. One of the four angles formed is 130 degrees. Find the measure of its ADJACENT angle and of the angle VERTICALLY OPPOSITE to the 130-degree angle. [\[Lines & Angles\]](#)

- (A) Adjacent = 230 degrees; opposite = 130 degrees
- (B) Adjacent = 60 degrees; opposite = 120 degrees
- (C) Adjacent = 50 degrees; opposite = 50 degrees
- (D) Adjacent = 50 degrees; vertically opposite = 130 degrees
- (E) Adjacent = 130 degrees; opposite = 130 degrees
- (F) Adjacent = 130 degrees; opposite = 50 degrees

13. A 350 mL volume of water percolates through a loamy soil column in 25 minutes. Expressed as a rate per hour, how much water would percolate in one hour at this steady rate? [\[Soil\]](#)

- (A) 875 mL per hour
- (B) 1400 mL per hour
- (C) 350 mL per hour
- (D) 14 mL per hour
- (E) 700 mL per hour
- (F) 375 mL per hour
- (G) 840 mL per hour

14. Express 64 as a power of 4, and also as a power of 2. Which single statement is fully correct? [\[Exponents & Powers\]](#)

- (A) $64 = 4^2 = 2^4$
- (B) $64 = 6^4 = 2^6$
- (C) $64 = 4^3 = 2^5$
- (D) $64 = 4^6 = 2^3$
- (E) $64 = 4^3 = 2^6$
- (F) $64 = 4^3 = 2^3$

15. A sea-breeze blows from sea to land during the day. Explain the chain: which surface heats faster, what happens to the air above it, and which way the breeze blows? [\[Heat\]](#)

- (A) Land stays coldest by day; no breeze forms
- (B) Land heats faster; warm air over land rises, cooler air from sea moves in - so the breeze blows from sea to land
- (C) Both heat equally; breeze blows randomly
- (D) Air over land sinks; breeze blows from land to sea
- (E) Land heats slower; cool air sinks over land, breeze blows land to sea
- (F) Sea cools faster by day; breeze blows from land

16. A transversal cuts two PARALLEL lines. One interior angle on a side is 3 times its co-interior (same-side interior) partner. Find the two co-interior angles. [\[Lines & Angles\]](#)

- (A) 40 degrees and 140 degrees
- (B) 50 degrees and 130 degrees
- (C) 45 degrees and 90 degrees
- (D) 60 degrees and 120 degrees
- (E) 30 degrees and 90 degrees
- (F) 45 degrees and 135 degrees
- (G) 36 degrees and 108 degrees

17. In a settling jar a gardener shakes soil with water; after resting, the sediment shows 6 cm of sand at the bottom, 4 cm of silt above it, and 2 cm of clay on top. What FRACTION of the total settled sediment is clay?

[\[Soil\]](#)

- (A) $\frac{1}{6}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{2}$
- (D) $\frac{2}{6}$ unsimplified
- (E) $\frac{1}{4}$
- (F) $\frac{2}{3}$
- (G) $\frac{1}{12}$

18. Use laws of exponents to simplify $(5^6 / 5^2) \times 5^0$ and give the value. [\[Exponents & Powers\]](#)

- (A) 625
- (B) 5^4 only as 20
- (C) 5^8
- (D) 1
- (E) 0
- (F) 125

19. A patient's temperature falls from 104 degF to 98.6 degF (normal). What is this fall expressed in degrees CELSIUS? Use $C = (5/9)(F - 32)$ for each reading. [\[Heat\]](#)

- (A) 37 degC
- (B) 3 degC
- (C) 5.4 degC
- (D) 40 degC
- (E) 9.72 degC
- (F) 2 degC
- (G) 6 degC

- 20.** Three angles meet at a point on one side of a straight line (a linear group on the line): they are x , $2x$ and $3x$ degrees. Find the value of the largest of these three angles. [\[Lines & Angles\]](#)
- (A) 60 degrees
 - (B) 30 degrees
 - (C) 120 degrees
 - (D) 45 degrees
 - (E) 90 degrees
 - (F) 180 degrees
- 21.** A loamy soil sample is 40% sand, 40% silt and the rest clay by mass. In a 2.5 kg dry sample, what is the mass of the clay portion? [\[Soil\]](#)
- (A) 100 g
 - (B) 1000 g
 - (C) 2000 g
 - (D) 20 g
 - (E) 250 g
 - (F) 500 g
 - (G) 200 g
- 22.** A sheet of paper is folded in half repeatedly. After each fold the number of layers doubles. Starting from 1 layer, how many layers are there after 10 folds? [\[Exponents & Powers\]](#)
- (A) 2048
 - (B) 10
 - (C) 1024
 - (D) 100
 - (E) 512
 - (F) 1000
 - (G) 20
- 23.** Why is the mercury (or alcohol) thread in a clinical thermometer made to stay put after the reading, with a kink near the bulb? Pick the fully correct reason. [\[Heat\]](#)
- (A) The kink is only decorative and has no use
 - (B) The kink heats the mercury further after removal
 - (C) The kink cools the bulb so the reading drops
 - (D) The kink turns the thermometer into a conductor
 - (E) The kink stops the liquid flowing back, so the reading holds until shaken down, letting the doctor read it after removal
 - (F) The kink makes the thermometer measure room temperature only
- 24.** An angle measures 64 degrees. State its complement and its supplement, then find the DIFFERENCE between the supplement and the complement. [\[Lines & Angles\]](#)
- (A) 64 degrees
 - (B) 116 degrees
 - (C) 128 degrees
 - (D) 154 degrees
 - (E) 26 degrees
 - (F) 90 degrees
 - (G) 52 degrees

25. Rainwater soaks fastest where particles are large and gaps are wide. A student ranks particle SIZE from largest to smallest. Which order is correct, and which soil therefore drains the quickest? [\[Soil\]](#)

- (A) Clay > sand > silt; clayey soil drains quickest
- (B) Sand > clay > silt; sandy soil drains quickest
- (C) Silt > clay > sand; silty soil drains quickest
- (D) Sand > silt > clay; sandy soil drains quickest
- (E) Sand = silt = clay; all drain equally
- (F) Clay > silt > sand; clayey soil drains quickest
- (G) Silt > sand > clay; silty soil drains quickest

26. Compare 3^4 and 4^3 by computing both. By how much is the larger one greater than the smaller?

[\[Exponents & Powers\]](#)

- (A) 0
- (B) 37
- (C) 17
- (D) 1
- (E) 7
- (F) 145
- (G) 27

27. A 2 cm gap is left between two 30 m steel rails to allow for expansion. In summer the temperature rises 25 degC and each rail expands by 0.5 mm per 10 degC. By how much does each rail expand, and is the 2 cm gap enough? [\[Heat\]](#)

- (A) Each rail expands 2.5 mm; the gap is too small
- (B) Each rail expands 1.25 cm; the gap is too small
- (C) Each rail expands 12.5 mm; the gap is just enough
- (D) Each rail expands 25 mm; the gap fails
- (E) Each rail expands 1.25 mm; the 20 mm gap is more than enough
- (F) Each rail expands 5 mm; gap exactly filled
- (G) Each rail expands 0.5 mm; gap unused

28. When a transversal crosses a pair of parallel rails, a matching pair of alternate interior angles work out to $(2x + 10)$ degrees and $(3x - 20)$ degrees. Find x and the size of each of these equal angles. [\[Lines & Angles\]](#)

- (A) $x = 30$; each angle = 70 degrees
- (B) $x = 30$; each angle = 130 degrees
- (C) $x = 30$; each angle = 90 degrees
- (D) $x = 2$; each angle = 14 degrees
- (E) $x = 10$; each angle = 30 degrees
- (F) $x = 30$; each angle = 50 degrees
- (G) $x = 6$; each angle = 22 degrees

29. A soil sample weighs 360 g when freshly dug (moist) and 288 g after being fully oven-dried. What percentage of the moist sample's mass was water? [\[Soil\]](#)

- (A) 72%
- (B) 12.5%
- (C) 20%
- (D) 25%
- (E) 28%
- (F) 80%
- (G) 18%

30. Write the prime factorisation of 360 in exponent form. Which is correct? [\[Exponents & Powers\]](#)

- (A) $2^2 \times 3^3 \times 5$
- (B) $2^3 \times 3^2 \times 5$
- (C) $2^2 \times 3^2 \times 5$
- (D) $2^3 \times 3^2 \times 5^2$
- (E) $2^3 \times 3 \times 5^2$
- (F) $3^3 \times 2^2 \times 5$

31. Two identical mugs of tea start at 80 degC. One is wrapped in wool, the other left bare, in a 20 degC room. After 10 minutes which is hotter, and which mode of heat loss does the wool mainly slow? [\[Heat\]](#)

- (A) The wrapped mug is hotter because wool generates heat
- (B) The wrapped mug is hotter; wool mainly slows conduction (and traps air to reduce heat loss)
- (C) Both are equal; wool does nothing
- (D) The bare mug is hotter; wool speeds cooling
- (E) The bare mug is hotter; wool adds heat
- (F) Both reach 20 degC instantly
- (G) The wrapped mug is cooler; wool conducts heat away fast

32. Around a single point several angles are drawn with no gaps: 110 degrees, 95 degrees, 65 degrees and one unknown angle. Find the unknown angle (angles around a point sum to 360 degrees). [\[Lines & Angles\]](#)

- (A) 100 degrees
- (B) 85 degrees
- (C) 270 degrees
- (D) 80 degrees
- (E) 90 degrees
- (F) 70 degrees
- (G) 110 degrees

33. Humus makes up 5% of a fertile topsoil by mass. A field's topsoil layer has a mass of 80,000 kg. After adding compost, the humus mass rises to 6,000 kg. By how many PERCENTAGE POINTS did the humus content increase? [\[Soil\]](#)

- (A) 5 percentage points
- (B) 3 percentage points
- (C) 1500 percentage points
- (D) 7.5 percentage points
- (E) 2.5 percentage points
- (F) 1 percentage point

- 34.** The mass of a grain of a substance is 3×10^{-3} g (i.e. 3 thousandths of a gram). What is the mass, in grams, of 2000 such grains, written in standard form? [\[Exponents & Powers\]](#)
- (A) 60 g
 - (B) 6×10^3 g
 - (C) 3×10^3 g
 - (D) 6 g
 - (E) 6×10^6 g
 - (F) 0.6 g
 - (G) 6×10^{-3} g
- 35.** A kitchen scale of temperatures: which list orders these from COLDEST to hottest - a freezer at -8 degC, melting ice at 0 degC, a warm room at 25 degC, and boiling water at 100 degC? [\[Heat\]](#)
- (A) 100 degC, 25 degC, 0 degC, -8 degC
 - (B) 0 degC, -8 degC, 25 degC, 100 degC
 - (C) -8 degC, 0 degC, 25 degC, 100 degC
 - (D) -8 degC, 25 degC, 0 degC, 100 degC
 - (E) 0 degC, 25 degC, 100 degC, -8 degC
 - (F) -8 degC, 0 degC, 100 degC, 25 degC
- 36.** Two supplementary angles differ by 36 degrees. Using a simple equation, find both angles. [\[Lines & Angles\]](#)
- (A) 72 degrees and 108 degrees
 - (B) 72 degrees and 144 degrees
 - (C) 90 degrees and 90 degrees
 - (D) 60 degrees and 96 degrees
 - (E) 82 degrees and 98 degrees
 - (F) 66 degrees and 114 degrees
- 37.** Field X lets 18 mm of rain soak in per hour; Field Y lets in 30 mm per hour. During a storm dropping 24 mm of rain in one hour, which field has surface run-off (water it cannot absorb), and how much? [\[Soil\]](#)
- (A) Field X, 12 mm of run-off
 - (B) Field Y, 6 mm of run-off
 - (C) Neither field has run-off
 - (D) Both fields, 6 mm each
 - (E) Field Y, 12 mm of run-off
 - (F) Field X, 6 mm of run-off
 - (G) Field X, 24 mm of run-off
- 38.** Simplify $(2^2 \times 3^2)$ and compare it with 6^2 . Which statement is true? [\[Exponents & Powers\]](#)
- (A) $2^2 \times 3^2 = 13$ and $6^2 = 36$
 - (B) $2^2 \times 3^2 = 72$
 - (C) $2^2 + 3^2 = 6^2$
 - (D) $2^2 \times 3^2 = 36$ but $6^2 = 12$
 - (E) $(2 \times 3)^2 = 2^2 \times 3^2 = 36$, so they are equal
 - (F) $2^2 \times 3^2 = 6^2 = 12$

- 39.** Heat moves three ways: conduction, convection and radiation. A person near a bonfire feels warmth even though air is a poor conductor and there is no wind toward them. Which mode mainly warms them, and why? [\[Heat\]](#)
- (A) Convection, because cold air sinks onto them
 - (B) Radiation, because heat travels as invisible rays straight to them without needing a medium
 - (C) Conduction, because the fire touches their skin
 - (D) None - they cannot feel the fire's heat
 - (E) Convection, because hot air blows directly at them
 - (F) Radiation, because the ground carries heat to their feet only
- 40.** A transversal makes a 75-degree angle with the first of two parallel lines. List the size of (a) the corresponding angle at the second line and (b) the co-interior angle at the second line. [\[Lines & Angles\]](#)
- (A) Corresponding = 90 degrees; co-interior = 90 degrees
 - (B) Corresponding = 15 degrees; co-interior = 165 degrees
 - (C) Corresponding = 105 degrees; co-interior = 105 degrees
 - (D) Corresponding = 75 degrees; co-interior = 105 degrees
 - (E) Corresponding = 105 degrees; co-interior = 75 degrees
 - (F) Corresponding = 75 degrees; co-interior = 75 degrees
 - (G) Corresponding = 75 degrees; co-interior = 25 degrees
- 41.** Paddy (rice) is grown in standing water, so it needs a soil that holds water and drains slowly. A student must pick the soil and justify it by particle size. Which choice is fully correct? [\[Soil\]](#)
- (A) Silty soil, because it cannot hold any water
 - (B) Clayey soil, because its tiny particles leave very small pores that hold water
 - (C) Sandy soil, because clay cannot grow crops
 - (D) Sandy soil, because its large particles hold water tightly
 - (E) Clayey soil, because its large particles drain water fastest
 - (F) Loamy soil, because it has the largest particles of all
 - (G) Sandy soil, because small pores drain slowly
- 42.** Bacteria double every 20 minutes. Starting from 50 bacteria, how many will there be after 2 hours, using powers of 2? [\[Exponents & Powers\]](#)
- (A) 3200
 - (B) 300
 - (C) 350
 - (D) 100
 - (E) 6400
 - (F) 1600
 - (G) 800
- 43.** A metal ball just passes through a metal ring at room temperature. The ball is then heated strongly. What happens when you try to pass it through the ring again, and why? [\[Heat\]](#)
- (A) It cannot be tested because metal does not conduct heat
 - (B) It passes more easily, because heat makes metal lighter
 - (C) It passes the same, because metals never change size
 - (D) It passes more easily, because heating shrinks the ball
 - (E) It melts the ring instantly
 - (F) It no longer passes, because the ball expands on heating and becomes larger
 - (G) It no longer passes, because the ring shrinks when the ball is hot

44. An angle is 4 times its own supplement's... no - an angle is such that it is 4 times its complement. Find the angle. [\[Lines & Angles\]](#)
- (A) 144 degrees
 - (B) 72 degrees
 - (C) 60 degrees
 - (D) 45 degrees
 - (E) 36 degrees
 - (F) 18 degrees
 - (G) 81 degrees
45. A dry clod of soil has mass 150 g. After soaking, it weighs 219 g, having absorbed water equal to part of its dry mass. What percentage of its dry mass did it absorb as water? [\[Soil\]](#)
- (A) 69%
 - (B) 46%
 - (C) 54%
 - (D) 23%
 - (E) 146%
 - (F) 31%
 - (G) 46.6 g
46. Evaluate $10^6 / 10^2$ and express how many times 10^2 fits into 10^6 . [\[Exponents & Powers\]](#)
- (A) 1000 times
 - (B) 100000 times
 - (C) 10 times
 - (D) 10^8 times
 - (E) 10^3 times
 - (F) 10000 times
47. An oven is set to 180 degC. Its thermostat keeps the inside within 5 degC of the set value. What is the highest temperature, in degrees Fahrenheit, the oven might reach? Use $F = (9/5)C + 32$. [\[Heat\]](#)
- (A) 365 degF
 - (B) 392 degF
 - (C) 356 degF
 - (D) 361 degF
 - (E) 185 degF
 - (F) 333 degF
48. At a point on a straight line, a ray makes two angles in the ratio 7 : 11 (a linear pair). Find the larger angle of the pair. [\[Lines & Angles\]](#)
- (A) 55 degrees
 - (B) 70 degrees
 - (C) 100 degrees
 - (D) 110 degrees
 - (E) 99 degrees
 - (F) 126 degrees
 - (G) 90 degrees

49. Weathering of rock into soil is extremely slow: about 1 cm of new soil forms in 200 years. At this rate, how many years are needed to build a 0.5 m deep soil layer? [\[Soil\]](#)

- (A) 5,000 years
- (B) 10,000 years
- (C) 1,000 years
- (D) 20,000 years
- (E) 2,000 years
- (F) 100 years

50. Simplify the expression $((2^3)^2 \times 2) / 2^3$, using the rule $(a^m)^n = a^{(mn)}$. [\[Exponents & Powers\]](#)

- (A) 2^6
- (B) 2^7
- (C) 2^2
- (D) 2^8
- (E) 2^4
- (F) 2^3
- (G) 16 written as 2^5

51. Why are the insides of houses in hot deserts often painted white, while people wear light-coloured clothes? Choose the complete reasoning. [\[Heat\]](#)

- (A) Light/white surfaces reflect most radiant heat and absorb little, keeping things cooler
- (B) Colour has no effect on heat at all
- (C) Dark surfaces reflect heat, so white is just for looks
- (D) Light clothes trap more body heat to keep you warm
- (E) White surfaces conduct heat fastest into the walls
- (F) White surfaces create their own cooling rays
- (G) White surfaces absorb the most heat, warming the house

52. Lines AB and CD intersect at O. Angle AOC = $(5x - 5)$ degrees and the vertically opposite angle BOD = $(3x + 35)$ degrees. Find x and the measure of angle AOC. [\[Lines & Angles\]](#)

- (A) $x = 20$; angle AOC = 95 degrees but BOD = 85
- (B) $x = 5$; angle AOC = 20 degrees
- (C) $x = 4$; angle AOC = 15 degrees
- (D) $x = 20$; angle AOC = 95 degrees
- (E) $x = 15$; angle AOC = 70 degrees
- (F) $x = 20$; angle AOC = 85 degrees

53. Three test pits show topsoil depths of 22 cm, 28 cm and 19 cm. A fourth pit's depth makes the average of all four exactly 25 cm. What is the depth at the fourth pit? [\[Soil\]](#)

- (A) 31 cm
- (B) 25 cm
- (C) 30 cm
- (D) 100 cm
- (E) 21 cm
- (F) 23 cm

54. A computer's storage is 2^{10} kilobytes (this equals 1 megabyte). A file is 2^6 kilobytes. How many such files fit in the storage? [\[Exponents & Powers\]](#)

- (A) 8 files
- (B) 4 files
- (C) 32 files
- (D) 16 files
- (E) 2^{16} files
- (F) 64 files
- (G) 1024 files

55. A 500 g block of metal needs 9000 J to rise by 20 degC. A 1.5 kg block of the SAME metal is given the same 20 degC rise. How much heat does the larger block need? [\[Heat\]](#)

- (A) 3000 J
- (B) 9000 J
- (C) 13500 J
- (D) 27000 J
- (E) 6000 J
- (F) 54000 J
- (G) 18000 J

56. An angle and its supplement are in the ratio 2 : 7. By how many degrees does the supplement EXCEED the angle? [\[Lines & Angles\]](#)

- (A) 140 degrees
- (B) 80 degrees
- (C) 20 degrees
- (D) 40 degrees
- (E) 90 degrees
- (F) 100 degrees
- (G) 120 degrees

57. An earthworm-rich plot loosens soil so percolation improves from 8 mL/min to 14 mL/min. Over a 90-minute watering, how much MORE water now percolates compared with before? [\[Soil\]](#)

- (A) 6 mL more
- (B) 60 mL more
- (C) 450 mL more
- (D) 540 mL less
- (E) 540 mL more
- (F) 720 mL more
- (G) 1260 mL more

58. Find the units digit of 2^{10} without writing every multiplication, by spotting the repeating pattern of units digits of powers of 2. [\[Exponents & Powers\]](#)

- (A) 6
- (B) 4
- (C) 8
- (D) 2
- (E) 5
- (F) 0
- (G) 1

59. A land breeze blows at night from land to sea. Put the cause in order: which surface cools faster at night, what the air above it does, and the resulting breeze direction. [\[Heat\]](#)

- (A) Air over land rises at night; breeze blows sea to land
- (B) Land cools faster; air over the warmer sea rises and cooler land air flows out to sea - a land-to-sea breeze
- (C) Both cool equally; no breeze at night
- (D) Land stays warmest at night; breeze blows sea to land
- (E) Land heats up at night; warm land air rises
- (F) Sea cools faster; land air sinks; breeze blows land to sea by chance
- (G) Sea cools faster; sea air rises; breeze blows sea to land

60. Two parallel lines are crossed by a transversal. A pair of co-interior angles are $(x + 40)$ degrees and $(2x + 20)$ degrees. Find x . [\[Lines & Angles\]](#)

- (A) $x = 80$
- (B) $x = 30$
- (C) $x = 60$
- (D) $x = 40$
- (E) $x = 45$
- (F) $x = 50$
- (G) $x = 20$

61. A rectangular plot is 40 m long and 25 m wide. To wet the topsoil, 12 litres of water are needed per square metre. How many KILOLITRES of water are needed to wet the whole plot once? (1 kilolitre = 1000 litres) [\[Soil\]](#)

- (A) 1000 kilolitres
- (B) 6 kilolitres
- (C) 12 kilolitres
- (D) 24 kilolitres
- (E) 1.2 kilolitres
- (F) 12000 kilolitres

62. Express the number 0.00072 in standard (scientific) form. [\[Exponents & Powers\]](#)

- (A) 72×10^{-5}
- (B) 7.2×10^{-4}
- (C) 7.2×10^4
- (D) 7.2×10^{-5}
- (E) 720×10^{-6} only
- (F) 0.72×10^{-3}

63. On the Celsius scale, the gap from the freezing point to the boiling point of water is divided into how many equal degrees, and what are those two fixed points? [\[Heat\]](#)

- (A) 98 equal degrees, between 0 and 98
- (B) 100 equal degrees, between -100 and 0
- (C) 100 equal degrees, between 0 degC (freezing) and 100 degC (boiling)
- (D) 212 equal degrees, between 0 and 212
- (E) 50 equal degrees, between 0 and 50
- (F) 100 equal degrees, between 32 and 132
- (G) 180 equal degrees, between 32 and 212

64. For a certain acute angle, its complement turns out to be exactly one-fourth of its supplement. Work out the size of this angle. [\[Lines & Angles\]](#)

- (A) 15 degrees
- (B) 75 degrees
- (C) 45 degrees
- (D) 36 degrees
- (E) 60 degrees
- (F) 30 degrees
- (G) 120 degrees

65. Soil erosion strips topsoil at 5 mm per year on a bare slope. The fertile topsoil layer is 15 cm deep. If nothing is done, after how many years will the fertile layer be completely gone? [\[Soil\]](#)

- (A) 45 years
- (B) 3 years
- (C) 300 years
- (D) 75 years
- (E) 20 years
- (F) 30 years
- (G) 150 years

66. A square has side 2^3 cm. Find its area as a power of 2, then state the area in ordinary numbers.

[\[Exponents & Powers\]](#)

- (A) $6^2 \text{ cm}^2 = 36 \text{ cm}^2$
- (B) $2^6 \text{ cm}^2 = 36 \text{ cm}^2$
- (C) $2^3 \text{ cm}^2 = 8 \text{ cm}^2$
- (D) $2^8 \text{ cm}^2 = 256 \text{ cm}^2$
- (E) $2^5 \text{ cm}^2 = 32 \text{ cm}^2$
- (F) $2^6 \text{ cm}^2 = 64 \text{ cm}^2$
- (G) $2^9 \text{ cm}^2 = 512 \text{ cm}^2$

67. A bimetallic strip of brass and iron is riveted together. Brass expands more than iron on heating. When heated, which way does the strip bend, and why is this useful? [\[Heat\]](#)

- (A) It bends toward the brass side and works as a magnet
- (B) It bends toward the brass side; useless for switches
- (C) It bends toward the iron side but has no practical use
- (D) It bends toward the iron side (brass on the outside of the curve); this makes it useful as an automatic switch in thermostats
- (E) It melts immediately on heating
- (F) It stays perfectly straight; metals expand equally
- (G) It curls into a full circle and snaps

68. Two angles form a linear pair. One is 24 degrees less than twice the other. Find the smaller angle. [\[Lines](#)

[& Angles\]](#)

- (A) 60 degrees
- (B) 44 degrees
- (C) 112 degrees
- (D) 68 degrees
- (E) 56 degrees
- (F) 78 degrees
- (G) 72 degrees

69. A nursery mixes garden soil and compost in the ratio 5 : 3 by mass to make 32 kg of potting mix. How many kilograms of COMPOST are in the mix? [\[Soil\]](#)

- (A) 12 kg
- (B) 18 kg
- (C) 10 kg
- (D) 8 kg
- (E) 15 kg
- (F) 20 kg
- (G) 3 kg

70. Which is larger: 2^5 or 5^2 , and what is their PRODUCT? [\[Exponents & Powers\]](#)

- (A) 2^5 is larger; product = 57
- (B) 2^5 is larger; product = 800
- (C) 5^2 is larger; product = 800
- (D) 2^5 is larger; product = 80
- (E) they are equal; product = 1024
- (F) 2^5 is larger; product = 8000
- (G) 5^2 is larger; product = 57

71. An ice cube at 0 degC is dropped into 200 g of water at 50 degC and the mixture settles at 10 degC, the ice having melted. Using 'heat lost by water = heat gained by ice', and 4 J per gram per degC for water, how much heat did the warm water lose? [\[Heat\]](#)

- (A) 16000 J
- (B) 32000 J
- (C) 4000 J
- (D) 2000 J
- (E) 8000 J
- (F) 800 J
- (G) 40000 J

72. A clock's hands at 4:00 form an angle at the centre. Knowing the 12 hour-marks divide 360 degrees equally, find this angle. [\[Lines & Angles\]](#)

- (A) 100 degrees
- (B) 60 degrees
- (C) 30 degrees
- (D) 150 degrees
- (E) 160 degrees
- (F) 120 degrees
- (G) 90 degrees

73. The layers of a soil profile from top to bottom are usually named in order. Which sequence is correct?

[Soil]

- (A) Bedrock, topsoil, weathered rock, subsoil
- (B) Topsoil (A-horizon), subsoil (B-horizon), weathered rock (C-horizon), bedrock
- (C) Subsoil, weathered rock, topsoil, bedrock
- (D) Topsoil, bedrock, subsoil, weathered rock
- (E) Bedrock, weathered rock, subsoil, topsoil
- (F) Weathered rock, bedrock, topsoil, subsoil
- (G) Subsoil, topsoil, bedrock, weathered rock

74. Simplify $(a^4 \times a^3) / (a^2 \times a)$ for a nonzero number a , giving the answer as a single power of a .

[Exponents & Powers]

- (A) a^3
- (B) a^2
- (C) a^{10}
- (D) a^4
- (E) a^7
- (F) a^5
- (G) a^{12}

75. Handles of cooking pans are made of plastic or wood, while the pan body is metal. State the property of each material and the reason for this choice. [Heat]

- (A) Wood conducts heat to keep the handle hot for safety
- (B) Both are equally good conductors; the choice is only for looks
- (C) Metal conducts heat well (to cook food); plastic/wood are poor conductors (insulators) so the handle stays cool to hold
- (D) Plastic conducts heat better than metal, so it cooks faster
- (E) Metal is an insulator; plastic conducts heat to the food
- (F) Metal is chosen for handles because it stays cool
- (G) Plastic is used on the body because it melts food evenly

76. Two parallel lines are cut by a transversal, creating eight angles. If one angle is 118 degrees, how many of the eight angles measure 118 degrees, and how many measure 62 degrees? [Lines & Angles]

- (A) Four are 118 degrees and four are 90 degrees
- (B) Six are 118 degrees and two are 62 degrees
- (C) Four angles are 118 degrees and four are 62 degrees
- (D) Three are 118 degrees and five are 62 degrees
- (E) Two are 118 degrees and six are 62 degrees
- (F) Five are 118 degrees and three are 62 degrees
- (G) All eight are 118 degrees

77. Two soils are tested for the same 300 mL of water. Soil P drains it in 12 min, Soil Q in 20 min. By what percentage is Soil P's percolation rate higher than Soil Q's? [\[Soil\]](#)

- (A) 40% higher
- (B) 166.7% higher
- (C) 8% higher
- (D) 20% higher
- (E) about 66.7% higher
- (F) 25% higher
- (G) 60% higher

78. A large number is 3.05×10^7 . Written in full, how many digits does this whole number have? [\[Exponents & Powers\]](#)

- (A) 7 digits
- (B) 5 digits
- (C) 8 digits
- (D) 6 digits
- (E) 9 digits
- (F) 305 digits
- (G) 10 digits

79. A weather station records midday maximum temperatures for six days: 31, 29, 34, 28, 33 and 27 (all degC). What is the MEAN of these six maximum readings? [\[Heat\]](#)

- (A) 30.33 degC (approx)
- (B) 33 degC
- (C) 182 degC
- (D) 28 degC
- (E) 29 degC
- (F) 30 degC
- (G) 31.2 degC

80. An angle is divided by a ray (bisected unequally) into two parts where one part is 28 degrees and the whole angle is supplementary to 110 degrees. Find the OTHER part of the angle. [\[Lines & Angles\]](#)

- (A) 152 degrees
- (B) 38 degrees
- (C) 42 degrees
- (D) 70 degrees
- (E) 82 degrees
- (F) 60 degrees
- (G) 52 degrees

81. A 1 hectare field ($10,000 \text{ m}^2$) loses topsoil to erosion. If 2 mm of soil is lost over the whole field and the soil's density is about 1.3 tonnes per cubic metre, what mass of soil is lost? (volume = area x depth) [\[Soil\]](#)

- (A) 52 tonnes
- (B) 13 tonnes
- (C) 2.6 tonnes
- (D) 26 tonnes
- (E) 20 tonnes
- (F) 2000 tonnes
- (G) 260 tonnes

- 82.** Evaluate the expression $2^4 + 2^4 + 2^4 + 2^4$ and write the answer as a single power of 2. [\[Exponents & Powers\]](#)
- (A) 2^5
 - (B) 2^6
 - (C) 2^{10}
 - (D) 2^8
 - (E) 2^{16}
 - (F) 8^4
 - (G) 4^4 as 2^4
- 83.** Water at the bottom of a beaker is heated. Coloured crystals show the water rises in the middle and sinks at the sides, forming a loop. Name this mode of heat transfer and the reason hot water rises. [\[Heat\]](#)
- (A) Convection; heated water expands, becomes less dense, and so rises while cooler denser water sinks
 - (B) Conduction; cold water rises because it is lighter
 - (C) Convection; hot water is denser and so rises
 - (D) Radiation; cold water sinks because it absorbs rays
 - (E) Convection; the beaker pushes water up mechanically
 - (F) Radiation; water carries rays upward
 - (G) Conduction; hot water is heavier and sinks
- 84.** Three angles of a linear group on a straight line are $(2x)$, $(3x + 10)$ and $(x + 50)$ degrees. Find x and the largest of the three angles. [\[Lines & Angles\]](#)
- (A) $x = 30$; largest = 100 degrees
 - (B) $x = 18$; largest = 64 degrees
 - (C) $x = 20$; largest = 40 degrees
 - (D) $x = 25$; largest = 85 degrees
 - (E) $x = 20$; largest = 70 degrees
 - (F) $x = 20$; largest = 60 degrees
 - (G) $x = 20$; largest = 80 degrees
- 85.** Loamy soil is best for most crops because it balances two opposite needs. Which pair of properties does loam combine well? [\[Soil\]](#)
- (A) Maximum water retention but no air spaces
 - (B) Only very large particles and no clay
 - (C) Maximum drainage but no water retention
 - (D) No humus but very large pebbles
 - (E) It cannot hold any nutrients at all
 - (F) Only very small particles and no sand
 - (G) Good water retention AND good drainage/aeration
- 86.** The distance light travels in one second is about 3×10^5 km. How far does it travel in 100 seconds, in standard form? [\[Exponents & Powers\]](#)
- (A) 3×10^3 km
 - (B) 30×10^6 km only
 - (C) 3×10^6 km
 - (D) 3×10^{500} km
 - (E) 3×10^5 km
 - (F) 3×10^7 km
 - (G) 300×10^5 km only

87. A fever of 102 degF is recorded. The doctor wants the rise above the normal 98.6 degF expressed in degrees Celsius. Convert each reading and find the rise. [\[Heat\]](#)

- (A) 6.1 degC
- (B) 2.4 degC
- (C) 38.9 degC
- (D) About 1.9 degC
- (E) 3.4 degC
- (F) 0.9 degC
- (G) 37 degC

88. The measure of an angle is 20 degrees more than its complement. Find the angle and confirm it is acute or obtuse. [\[Lines & Angles\]](#)

- (A) 110 degrees, which is obtuse
- (B) 45 degrees, which is acute
- (C) 65 degrees, which is acute
- (D) 55 degrees, which is acute
- (E) 35 degrees, which is acute
- (F) 80 degrees, which is acute

89. A percolation test gives 16 mL/min for sandy soil. A field of this soil is watered for 1 hour 15 minutes. Assuming the rate stays steady, how many LITRES of water percolate? (1 L = 1000 mL) [\[Soil\]](#)

- (A) 1.2 mL
- (B) 75 litres
- (C) 0.96 litres
- (D) 1200 litres
- (E) 1.2 litres
- (F) 12 litres

90. Simplify $(7^0 + 3^2) \times 2^3$ using the order of operations and the rule that any nonzero number to the power 0 is 1. [\[Exponents & Powers\]](#)

- (A) 72
- (B) 640
- (C) 18
- (D) 80
- (E) 26
- (F) 800
- (G) 81

91. Why does a sealed glass bottle filled to the brim with water crack if left in a freezer? Choose the complete explanation. [\[Heat\]](#)

- (A) Cold makes glass weightless so it breaks
- (B) Water neither expands nor contracts; the break is random
- (C) The freezer adds water that overflows and breaks it
- (D) Water is unusual: it EXPANDS as it freezes into ice, and the trapped expansion pushes the glass until it cracks
- (E) Water shrinks when freezing, creating a vacuum that implodes the bottle
- (F) The glass expands more than the ice and shatters itself
- (G) Ice conducts heat into the glass and melts it

92. Two angles are both supplementary to the SAME 118-degree angle (one on each side). What can you conclude about these two angles, and what is each one's measure? [\[Lines & Angles\]](#)

- (A) They are equal, each 32 degrees
- (B) They are equal, each 72 degrees
- (C) One is 62 and the other is 28 degrees
- (D) They are equal, each measuring 62 degrees
- (E) They add to 90, so each is 45 degrees
- (F) They differ, one 62 and one 118 degrees
- (G) They are equal, each 118 degrees

93. In a soil sample of 500 g, the air-filled pore space is found to be 30% of the soil's bulk volume, while particles fill the remaining 70%. If the bulk volume is 400 cm^3 , what volume is taken up by air-filled pores?

[\[Soil\]](#)

- (A) 70 cm^3
- (B) 350 cm^3
- (C) 120 cm^3
- (D) 30 cm^3
- (E) 120 g
- (F) 280 cm^3

94. A chessboard trick: 1 grain on the first square, then doubling each square. How many grains are on the 8th square alone? [\[Exponents & Powers\]](#)

- (A) 128
- (B) 16
- (C) 256
- (D) 255
- (E) 127
- (F) 8

95. A solar cooker reaches 90 degC inside on a clear day. A recipe needs the food held above 85 degC for 30 minutes, but each passing cloud drops the inside by 12 degC for as long as it covers the Sun. During a cloud, is the food still above 85 degC , and by how much does it miss or clear the mark? [\[Heat\]](#)

- (A) No; it drops to 81 degC , 4 degC below
- (B) Yes; it stays at 90 degC unchanged
- (C) Yes; it stays at 88 degC , 3 degC above
- (D) Yes; it rises to 102 degC
- (E) No; it drops to 73 degC , 12 degC below
- (F) No; it drops to 78 degC , which is 7 degC below the 85 degC needed

96. A ray stands on a line making a linear pair. The ratio of the two angles is 1 : 5. By how many degrees is the larger angle GREATER than a right angle? [\[Lines & Angles\]](#)

- (A) 150 degrees
- (B) 75 degrees
- (C) 60 degrees
- (D) 120 degrees
- (E) 45 degrees
- (F) 30 degrees
- (G) 90 degrees

97. Adding lime can change soil, but here a student lists processes that FORM soil from rock. Which list contains ONLY soil-forming (weathering) processes? [\[Soil\]](#)

- (A) Magnetising rocks and burning fossil fuels
- (B) Cracking by heat and cold, breaking by flowing water, and action of roots and lichens
- (C) Boiling of underground water and freezing of oceans
- (D) Evaporation of seawater and condensation of clouds
- (E) Photosynthesis in leaves and respiration in animals
- (F) Melting of rock into lava and cooling underground only

98. Find the value of $5^3 - 5^2$ and express the result as a multiple of a power of 5. [\[Exponents & Powers\]](#)

- (A) 100, which equals 4×5^2
- (B) 625, which equals 5^4
- (C) 100, which equals 5×5^2
- (D) 150, which equals 6×5^2
- (E) 25, which equals 5^2
- (F) 20, which equals 4×5
- (G) 100, which equals 5^3

99. Rank these materials by how WELL they conduct heat, from best to worst: copper, water, wood, still air. [\[Heat\]](#)

- (A) Copper (best), water, wood, still air (worst)
- (B) Copper, wood, water, air
- (C) Water, copper, air, wood
- (D) Copper, air, water, wood
- (E) Air, water, wood, copper
- (F) Wood, copper, air, water

100. Two parallel lines are cut by a transversal. The bisector of one interior angle and the bisector of its co-interior partner meet. Given the two co-interior angles are 80 and 100 degrees, find the sum of their HALF-angles (the two bisected halves). [\[Lines & Angles\]](#)

- (A) 80 degrees
- (B) 60 degrees
- (C) 45 degrees
- (D) 180 degrees
- (E) 50 degrees
- (F) 100 degrees
- (G) 90 degrees